

Transportation Telemetry Predictive Timing Platform

Synthetic train telemetry, lakehouse pipeline, and neural network delay prediction

Project question

Can a real-time transportation telemetry platform predict train delay timing and support predictive maintenance decision-making using sensor, location, and operational event data?

[Overview](#) [Metrics](#) [Figures](#) [Architecture](#)

This project demonstrates:

- Bronze / Silver / Gold lakehouse pipeline
- synthetic train telemetry events
- neural network predictive timing model
- delay risk classification
- delay minutes regression
- portfolio-ready visual outputs